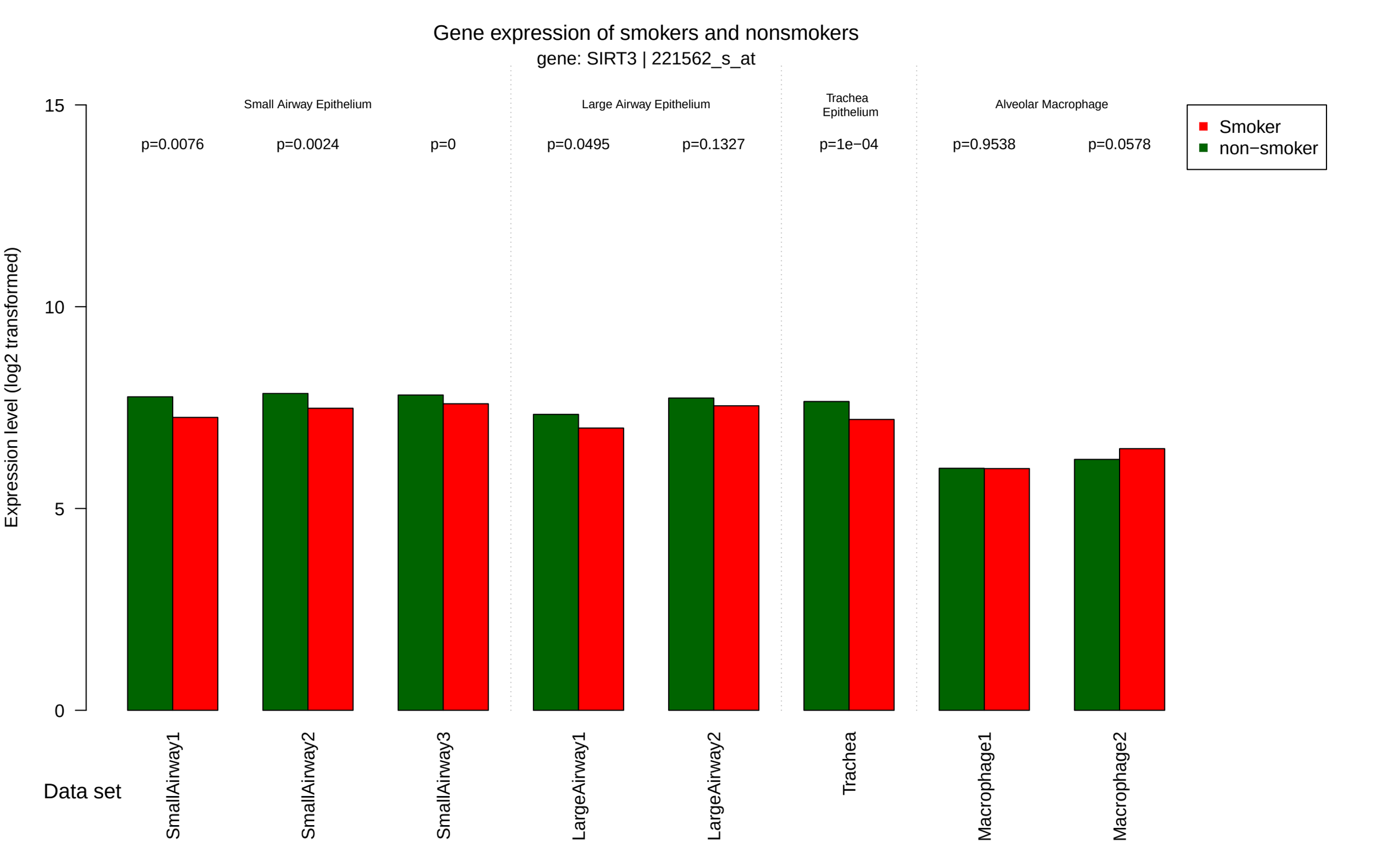
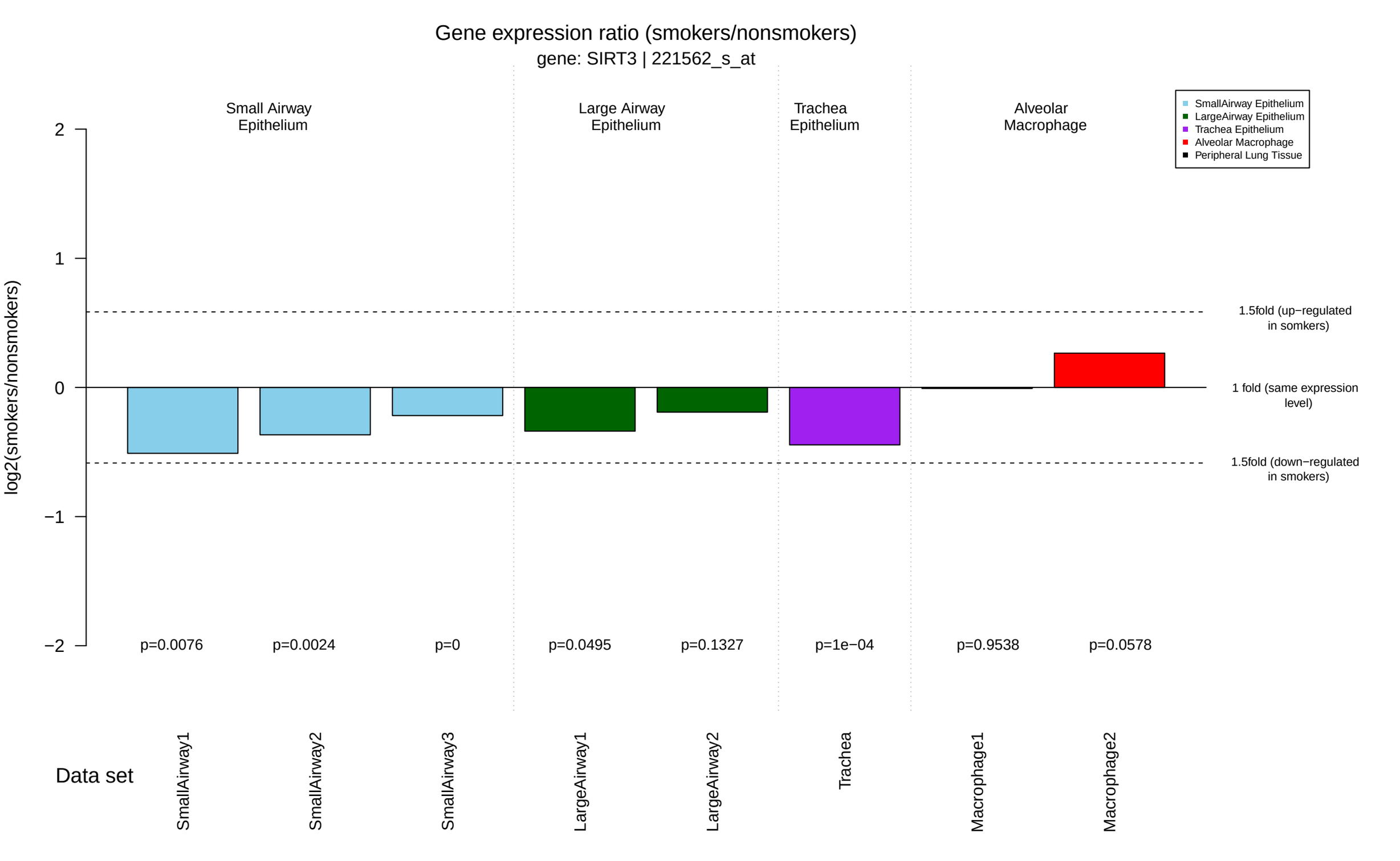
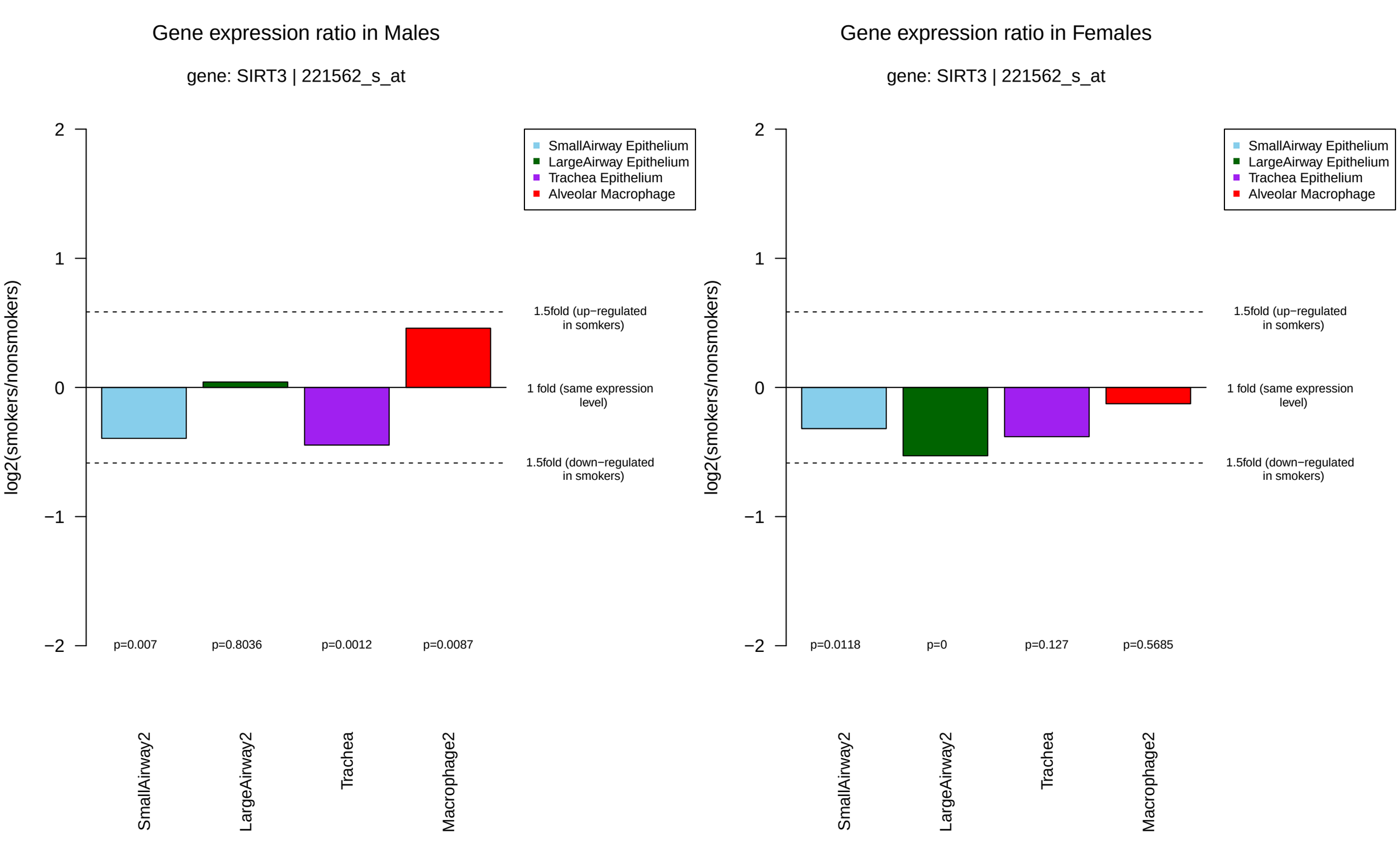
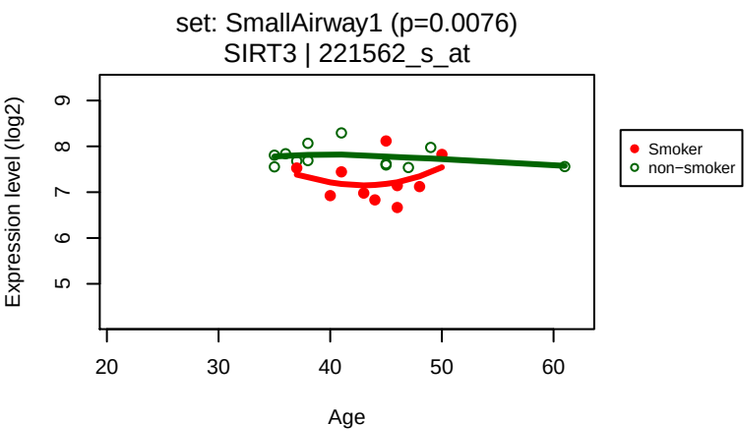
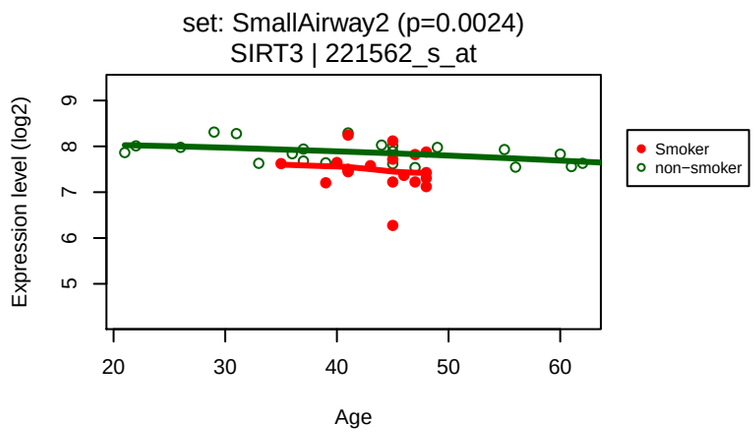
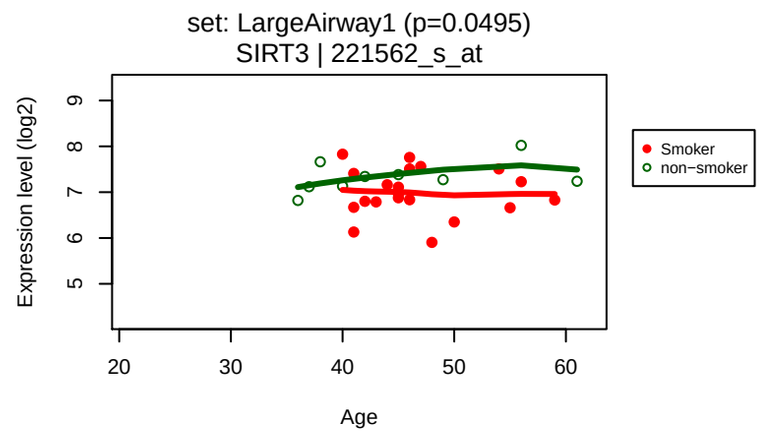

Gene expression of smokers and nonsmokers
gene: SIRT3 | 221562_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0076
p=0.0024
p=0
p=0.0495
p=0.1327
p=1e−04
p=0.9538
p=0.0578
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

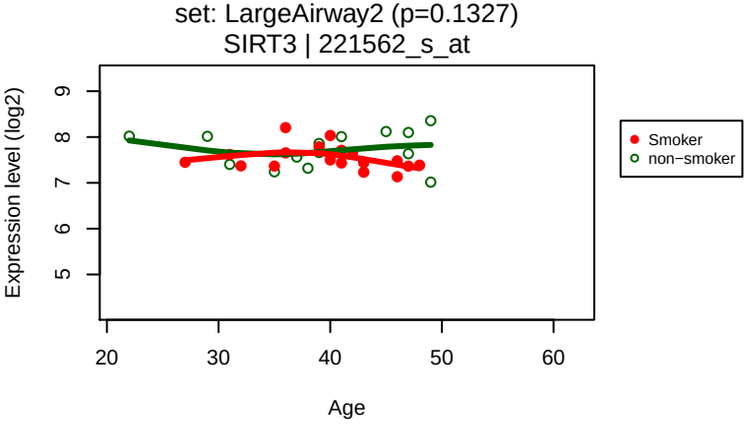

Gene expression ratio (smokers/nonsmokers)
gene: SIRT3 | 221562_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0076
p=0.0024
p=0
p=0.0495
p=0.1327
p=1e−04
p=0.9538
p=0.0578
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

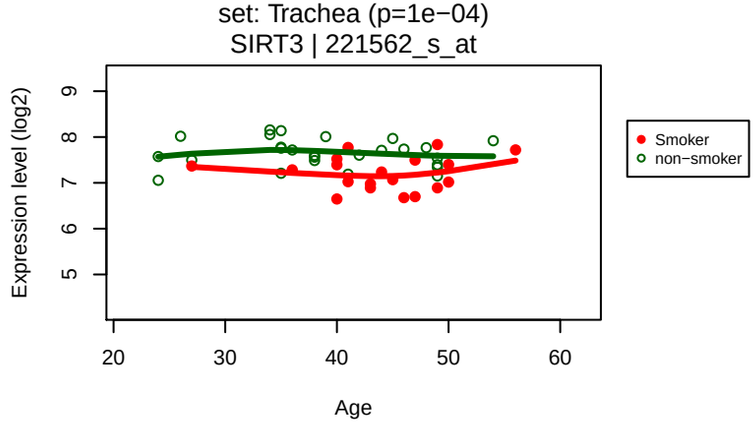

Gene expression ratio in Males
gene: SIRT3 | 221562_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.007
p=0.8036
p=0.0012
p=0.0087
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SIRT3 | 221562_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0118
p=0
p=0.127
p=0.5685
SmallAirway2
LargeAirway2
Trachea
Macrophage2

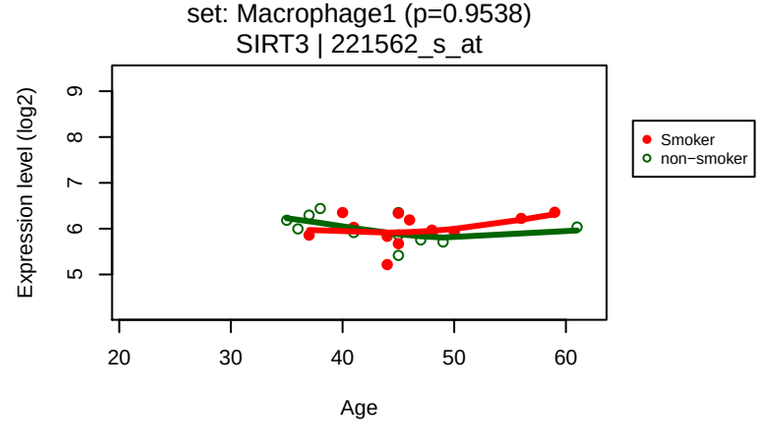

set: SmallAirway1 (p=0.0076)
SIRT3 | 221562_s_at
Expression level (log2)
Age
Smoker
non−smoker

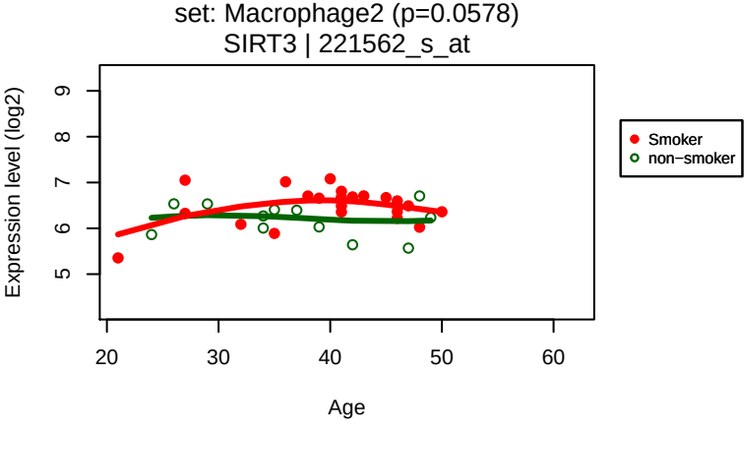

set: SmallAirway2 (p=0.0024)
SIRT3 | 221562_s_at
Expression level (log2)
Age
Smoker
non−smoker

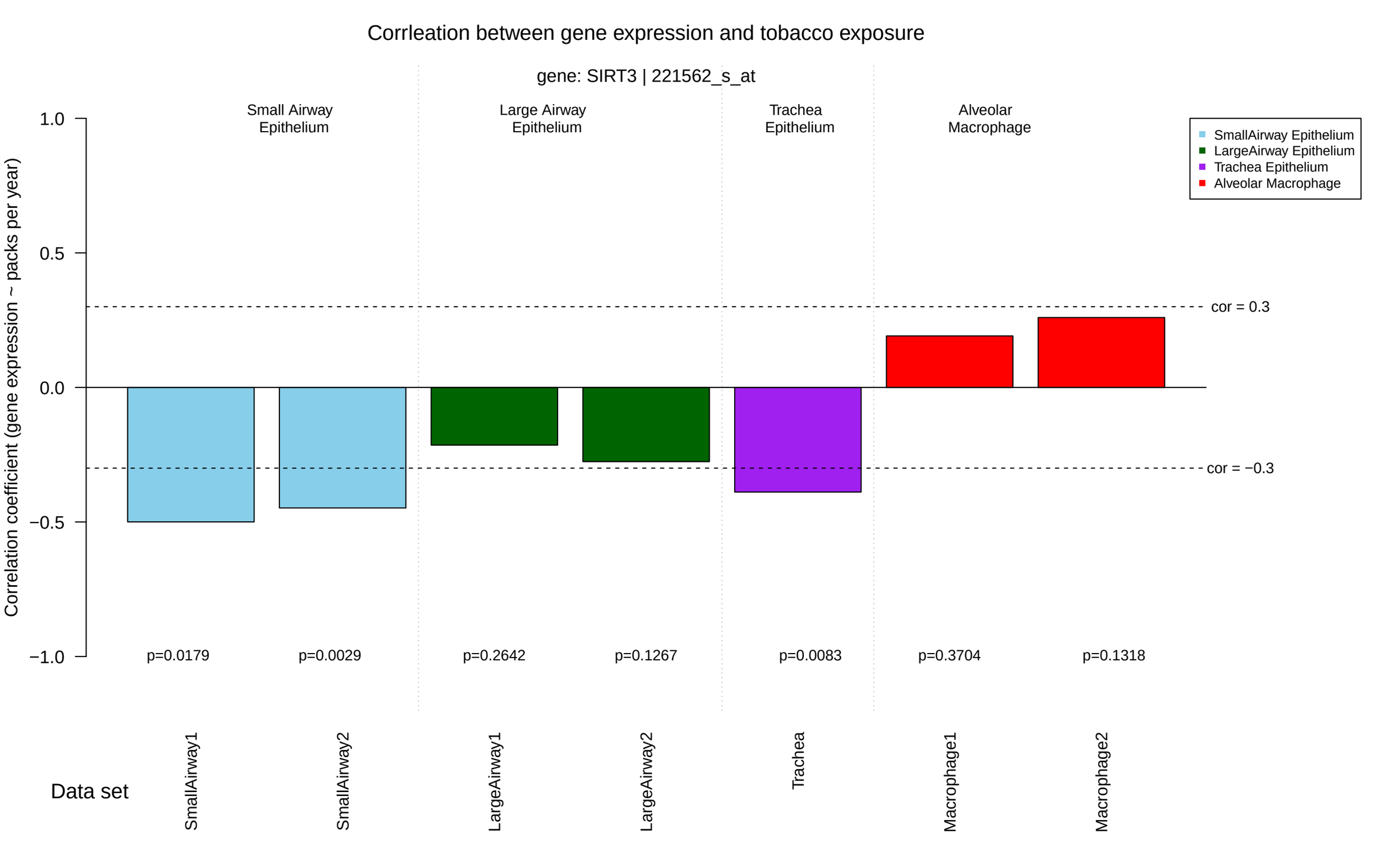

set: LargeAirway1 (p=0.0495)
SIRT3 | 221562_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1327)
SIRT3 | 221562_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=1e−04)
SIRT3 | 221562_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9538)
SIRT3 | 221562_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0578)
SIRT3 | 221562_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SIRT3 | 221562_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0179
p=0.0029
p=0.2642
p=0.1267
p=0.0083
p=0.3704
p=0.1318
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2